

Development of local analysis for program equivalence

(How I spent five years to get papers published)

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(Project researcher @ NII, Japan)

Special thanks to... Kazunori Ueda (Waseda Univ.), Makoto Hamana (Kyusyu Institute of Technology),
Shigeru Chiba (Univ. Tokyo), Ichiro Hasuo (NII)
Natsuki Urabe (NII), Takahiro Sanada (Fukui Prefectural Univ.)

Modelling program execution

- software science
 - trustable software
 - verification of systems that execute programs

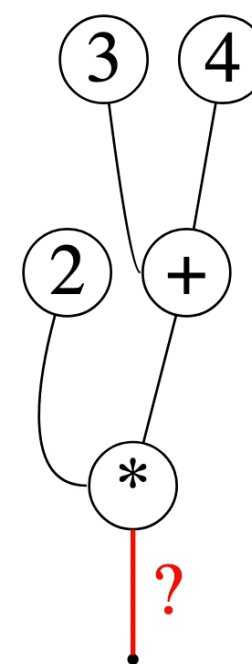
Behavioural equivalence between programs

- software science
 - trustable software
 - verification of systems that execute programs
 - mathematical proof of behavioural equivalence between programs

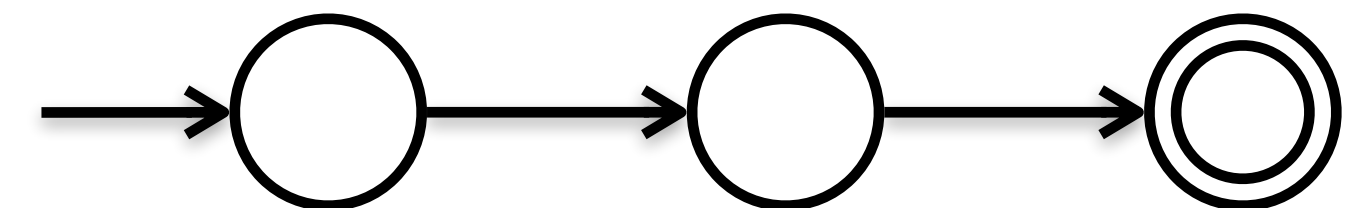
term rewriting system

$$2 \times (3 + 4) \rightarrow 2 \times 7 \rightarrow 14$$

graph rewriting system



(nondeterministic) automaton



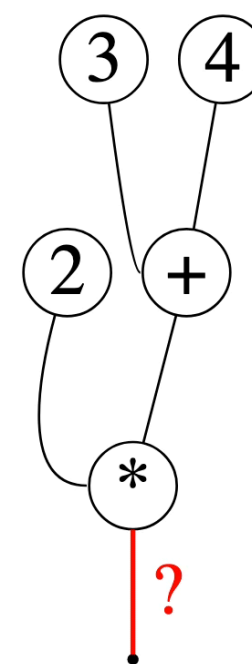
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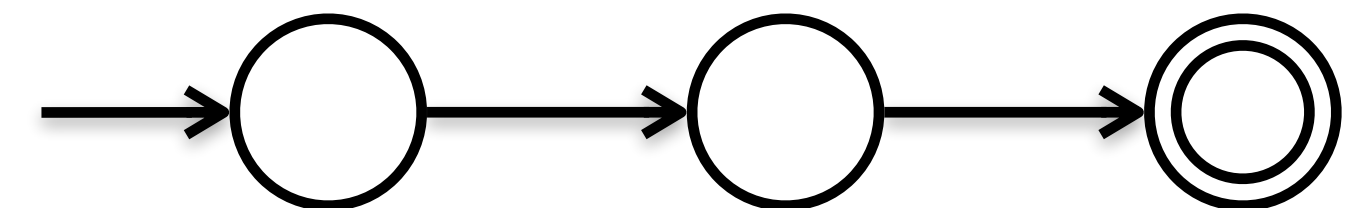
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Behavioural equivalence between programs

- model checking: $\mathcal{S} \models \phi$

system

specification

- behavioural equivalence: $P \cong P' \stackrel{\text{def.}}{\iff} \forall \phi . (\mathcal{S}_P \models \phi \iff \mathcal{S}_{P'} \models \phi)$

system
specialised to
program P

system
specialised to
program P'

...typically: $P \cong P' \stackrel{\text{def.}}{\iff} \forall r . (\mathcal{S}_P \models \downarrow \{r\} \iff \mathcal{S}_{P'} \models \downarrow \{r\})$

termination
with result r

Behavioural equivalence between programs

- a mathematical proof of behavioural equivalence
 $\implies$ certification that two (versions of) programs “behave the same”
- proof methodologies
 - established:
applicative bisimulation [Abramsky '90], logical relation [Plotkin '73] etc.
using term rewriting systems

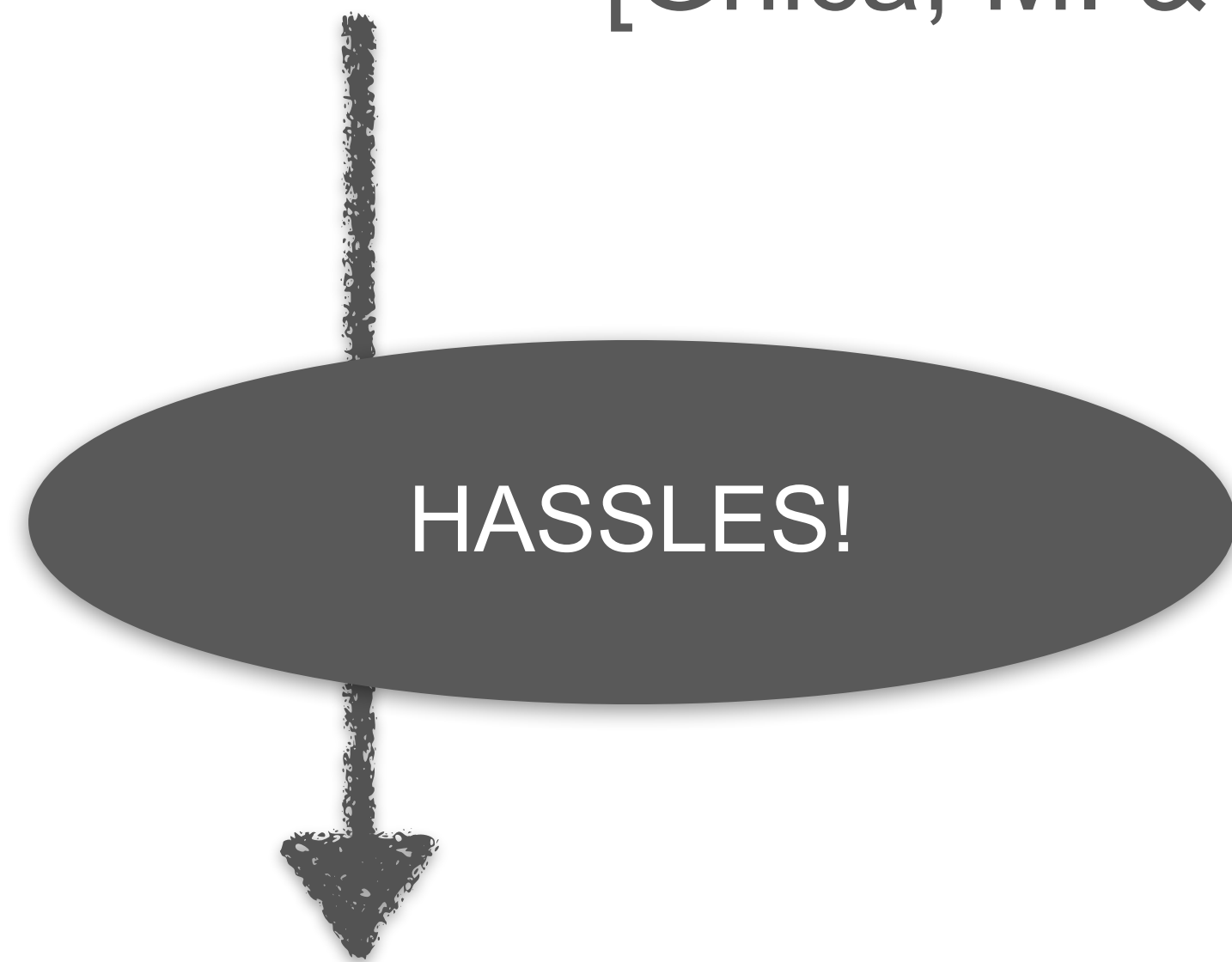
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using term rewriting systems
 - radically new: *local analysis* using graph rewriting systems

Too long to get a result published...

- **Result** *local analysis* using graph rewriting systems

[Ghica, M. & Waugh Ambridge, arxiv 2019] [M., PhD thesis 2020]



- **Paper** [M., Sanada & Urabe, CMCS 2024]
- **Paper** [M. & Hamana, FLOPS 2024]

Learned to *carry* research

- **Motivation** fragile proof methodologies of behavioural equivalence
- **Approach** use graph rewriting systems, instead of term rewriting systems
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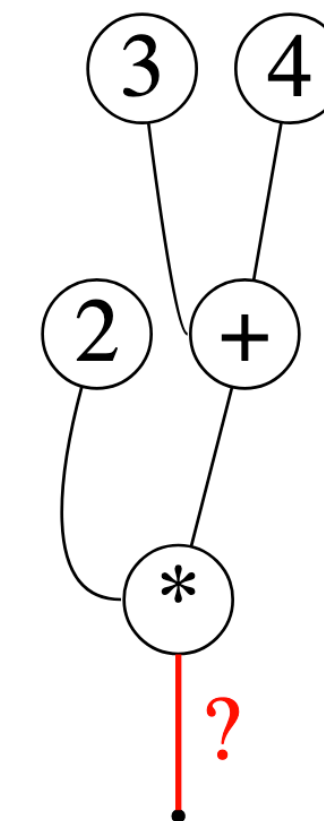
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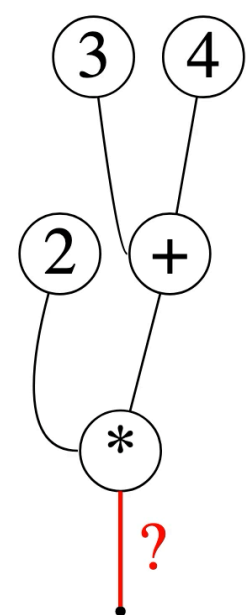
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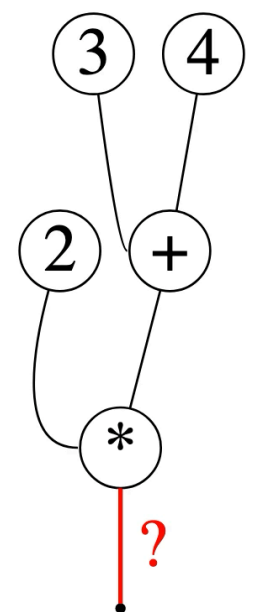
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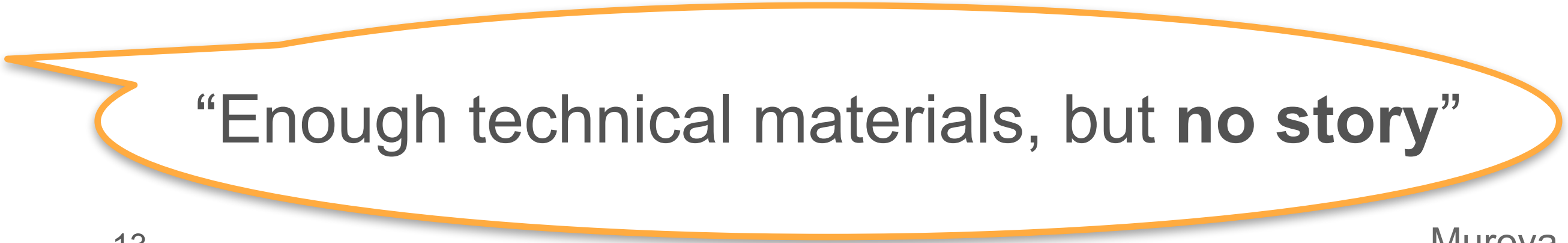
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Struggled even more...

- COVID-19
- a mental health issue: depression
 - I was appointed as an assistant professor, overwhelmed by freedom.
 - A collaborator abruptly quitted research.
 - I strongly felt I had to stand on my own two feet.
 - having little chance of talking to my colleagues...

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“What is your work in a nutshell?”

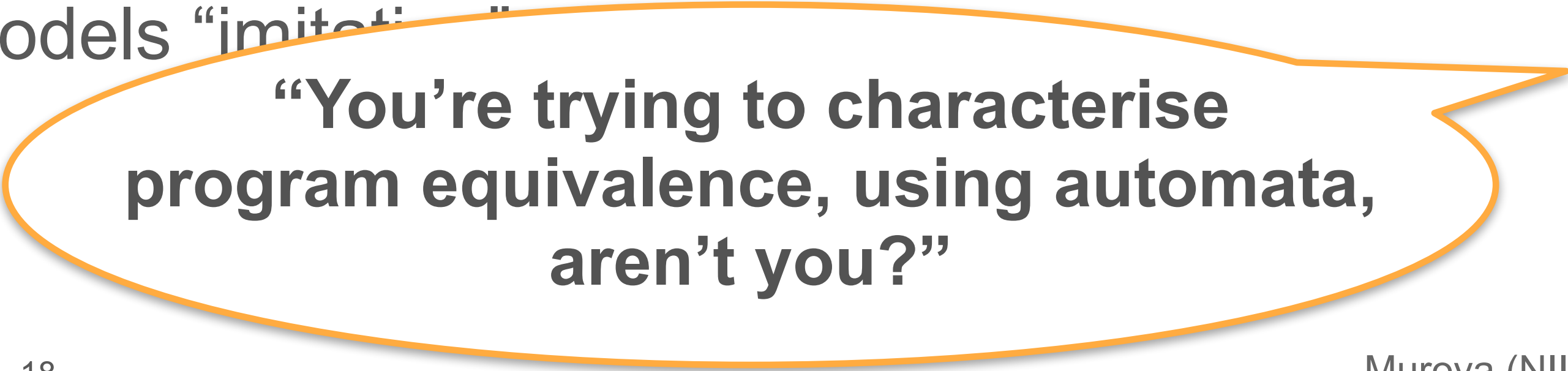
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“You’re trying to characterise
program equivalence, using automata,
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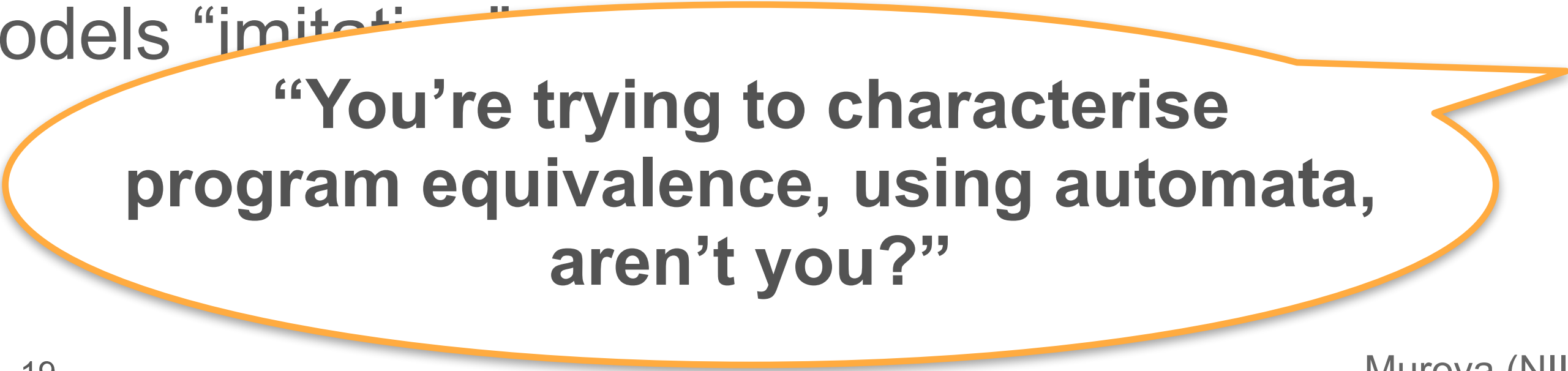
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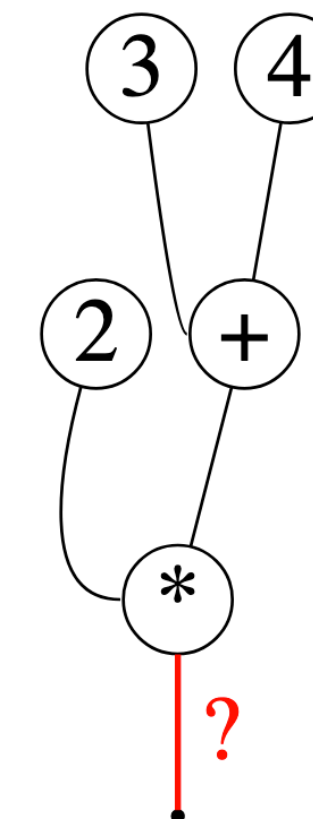


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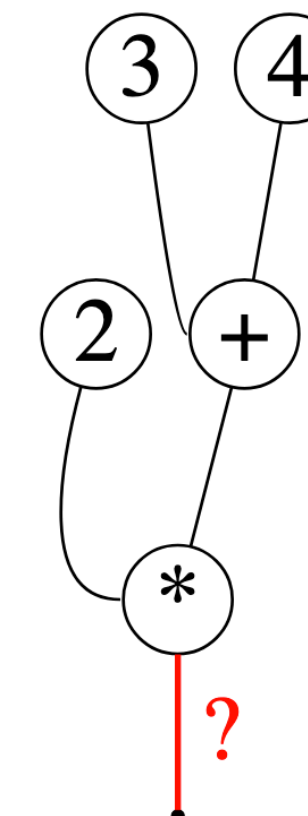


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- **Approach 1'** use term rewriting systems directly

2 × (3 + 4)



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- **Approach 1'** use term rewriting systems directly
- **Motivation** no *evaluation strategies* in term rewriting systems
- **Problem** *evaluation contexts* in term rewriting systems

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- **Problem** *evaluation contexts* in term rewriting systems
- **Solution** *term evaluation systems*
- **Paper** [M. & Hamana, FLOPS 2024]



Too long to get a result published... but worth it (I hope)

- **Result** *local analysis* using graph rewriting systems

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HASSLES!

- ✓ • **Paper** [M., Sanada & Urabe, CMCS 2024]
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To *frame* research

How I thought of research:

- **Motivation** (why you care)
- **Approach** (how you tackle challenges)
- **Result** (what you obtain)

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How I think research should be framed:

- **Motivation** (why people should care)
- **Approach** (how you tackle challenges)
- **Solution** (what you offer)

Key (1)

take into account your research
community's perspective

To *frame* research

How I think research should be framed:

- **Motivation** (why people should care)
- **Problem** (what your solution is for)
- **Approach** (how you tackle challenges)
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Key (1)

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Key (2)

clarify what your solution can do
and what it cannot